Counting

Chapter 6

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Chapter Summary

6.1 The Basics of Counting

6.2 The Pigeonhole Principle

6.3 Permutations and Combinations

6.4 Binomial Coefficients and Identities

6.5 Generalized Permutations and Combinations

6.6 Generating Permutations and Combinations

The Basics of Counting

Section 6.1

6.1.1 Introduction

Suppose that a password on a computer system consists of six, seven, or eight characters.

Each of these characters must be a digit or a letter of the alphabet.

Each password must contain at least one digit.

How many such passwords are there?

The techniques needed to answer this question and a wide variety of other counting problems will be introduced in this section.

- Product Rule
- Sum Rule
- Subtraction Rule
- Division Rule

6.1.2 Basic Counting Principles

Product Rule

Sum Rule

The Product Rule

The Product Rule: A procedure can be broken down into a sequence of two tasks. There are n_1 ways to do the first task and n_2 ways to do the second task. Then there are $n_1 \cdot n_2$ ways to do the procedure.

Example: How many bit strings of length seven are there?

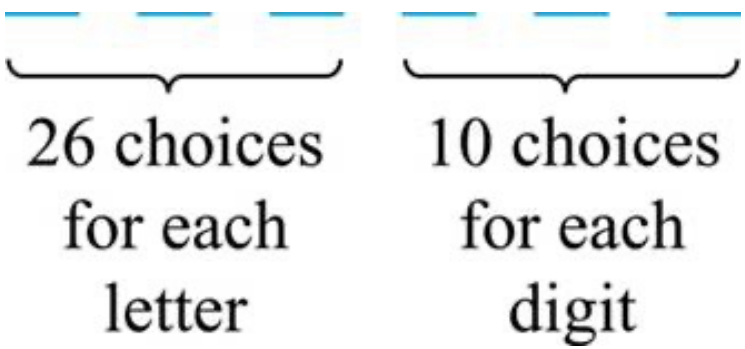
Solution: Since each of the seven bits is either a 0 or a 1, the answer is $2^7 = 128$.

The Product Rule

Example: How many different license plates can be made if each plate contains a sequence of three uppercase English letters followed by three digits?

Solution: By the product rule,

there are $26 \cdot 26 \cdot 26 \cdot 10 \cdot 10 \cdot 10 = 17{,}576{,}000$ different possible license plates.



Counting Functions

Counting Functions: How many functions are there from a set with m elements to a set with n elements?

Solution: Since a function represents a choice of one of the n elements of the codomain for each of the m elements in the domain, the product rule tells us that there are $n \cdot n \cdots n = n^m$ such functions.

Counting One-to-One Functions: How many one-to-one functions are there from a set with m elements to one with n elements (m<=n)?

Solution: Suppose the elements in the domain are $a_1, a_2, \ldots, a_m$. There are n ways to choose the value of a_1 and $n-1$ ways to choose a_2, etc. The product rule tells us that there are $n(n-1)(n-2)\cdots(n-m+1)$ such functions.

Telephone Numbering Plan

Example: The *North American numbering plan* (*NANP*) specifies that a telephone number consists of 10 digits, consisting of a three-digit area code, a three-digit office code, and a four-digit station code. There are some restrictions on the digits.

- Let *X* denote a digit from 0 through 9.
- Let *N* denote a digit from 2 through 9.
- Let *Y* denote a digit that is 0 or 1.
- In the old plan (in use in the 1960s) the format was *NYX-NNX-XXXX*.
- In the new plan, the format is *NXX-NXX-XXXX*.

How many different telephone numbers are possible under the old plan and the new plan?

Solution: Use the Product Rule.

- There are $8 \cdot 2 \cdot 10 = 160$ area codes with the format *NYX*.
- There are $8 \cdot 10 \cdot 10 = 800$ area codes with the format *NXX*.
- There are $8 \cdot 8 \cdot 10 = 640$ office codes with the format *NNX*.
- There are $10 \cdot 10 \cdot 10 \cdot 10 = 10{,}000$ station codes with the format *XXXX*.

Number of old plan telephone numbers: $160 \cdot 640 \cdot 10{,}000 = 1{,}024{,}000{,}000$.

Number of new plan telephone numbers: $800 \cdot 800 \cdot 10{,}000 = 6{,}400{,}000{,}000$.

Counting Functions

Example: What is the value of k after the following code, where n_1, n_2, … , n_m are positive integers, has been executed?

```
k := 0
for i_1 := 1 to n_1
    for i_2 := 1 to n_2
        .
        .
        .
        for i_m := 1 to n_m
            k := k + 1
```

$n_1 n_2 \cdots n_m$

Counting Subsets of a Finite Set

Use the product rule to show that the number of different subsets of a finite set S is $2^{|S|}$.

Solution:

When the elements of S are listed in an arbitrary order, there is a one-to-one correspondence between subsets of S and bit strings of length $|S|$.

When the ith element is in the subset, the bit string has a 1 in the ith position and a 0 otherwise.

By the product rule, there are $2^{|S|}$ such bit strings, and therefore $2^{|S|}$ subsets.

Product Rule in Terms of Sets

If $A_1, A_2, \ldots, A_m$ are finite sets, then the number of elements in the Cartesian product of these sets <u>is</u> the product of the number of elements of each set.

The task of <u>choosing an element</u> in the Cartesian product $A_1 \times A_2 \times \cdots \times A_m$ is done by <u>*choosing an element*</u> in A_1, an element in A_2 , …, and an element in A_m.

By the product rule, it follows that:

$$|A_1 \times A_2 \times \cdots \times A_m| = |A_1| \cdot |A_2| \cdot \cdots \cdot |A_m|$$

The Sum Rule

The Sum Rule: If a task can be done either in one of n_1 ways or in one of n_2, where none of the set of n_1 ways is the same as any of the n_2 ways, then there are $n_1 + n_2$ ways to do the task.

Example: The mathematics department must choose either a student or a faculty member as a representative for a university committee. How many choices are there for this representative if there are 37 members of the mathematics faculty and 83 mathematics majors and no one is both a faculty member and a student.

Solution: By the sum rule it follows that there are 37 + 83 = 120 possible ways to pick a representative.

The Sum Rule

Example: What is the value of k after the following code, where n_1, n_2, … , n_m are positive integers, has been executed?

```
k := 0
for i_1 := 1 to n_1
    k := k + 1
for i_2 := 1 to n_2
    k := k + 1
        .
        .
        .
for i_m := 1 to n_m
    k := k + 1
```

$n_1 + n_2 + \cdots + n_m$

The Sum Rule in terms of sets.

The sum rule can be phrased in terms of sets.

$|A \cup B| = |A| + |B|$ as long as A and B are disjoint sets.

Or more generally,

$$|A_1 \cup A_2 \cup \cdots \cup A_m| = |A_1| + |A_2| + \cdots + |A_m| \text{ when } A_i \cap A_j = \emptyset \text{ for all } i, j.$$

6.1.3 More Complex Counting Problems

Many counting problems cannot be solved using just the sum rule or just the product rule.

However, many complicated counting problems can be solved using both of these rules in combination.

Combining the Sum and Product Rule.

Example: Suppose statement labels in a programming language can be either a single letter or a letter followed by a digit. Find the number of possible labels.

Solution: Use the product rule.

$$26 + 26 \cdot 10 = 286$$

Counting Passwords

Example: Each user on a computer system has a password, which is six to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there?

Solution: Let P be the total number of passwords, and let P_6, P_7, and P_8 be the passwords of length 6, 7, and 8. By the sum rule $P = P_6 + P_7 + P_8$.

- $P_6 = 36^6 - 26^6$ =2,176,782,336 − 308,915,776 =1,867,866,560.
- $P_7 = 36^7 - 26^7$ =78,364,164,096 − 8,031,810,176 = 70,332,353,920.
- $P_8 = 36^8 - 26^8$ =2,821,109,907,456 − 208,827,064,576 = 2,612,282,842,880.

Consequently, $P = P_6 + P_7 + P_8 = 2{,}684{,}483{,}063{,}360$.

Internet Addresses

Version 4 of the Internet Protocol (IPv4) uses 32 bits.

Bit Number	0	1	2	3	4	8	16	24	31
Class A	0	netid				hostid			
Class B	1	0	netid				hostid		
Class C	1	1	0	netid				hostid	
Class D	1	1	1	0	Multicast Address				
Class E	1	1	1	1	0	Address			

Class A Addresses: used for the largest networks, a 0,followed by a 7-bit netid and a 24-bit hostid.

Class B Addresses: used for the medium-sized networks, a 10,followed by a 14-bit netid and a 16-bit hostid.

Class C Addresses: used for the smallest networks, a 110,followed by a 21-bit netid and a 8-bit hostid.

- Neither Class D nor Class E addresses are assigned as the address of a computer on the internet. Only Classes A, B, and C are available.
- 1111111 is not available as the netid of a Class A network.
- Hostids consisting of all 0s and all 1s are not available in any network.

Counting Internet Addresses

Example: How many different IPv4 addresses are available for computers on the internet?

Bit Number	0	1	2	3	4	8	16	24	31
Class A	0	netid				hostid			
Class B	1	0	netid				hostid		
Class C	1	1	0	netid				hostid	
Class D	1	1	1	0	Multicast Address				
Class E	1	1	1	1	0	Address			

Not Enough Today !!
The newer IPv6 protocol solves the problem of too few addresses.

Solution: Use both the sum and the product rule. Let x be the number of available addresses, and let x_A, x_B, and x_C denote the number of addresses for the respective classes.

- To find, x_A: $2^7 - 1 = 127$ netids. $2^{24} - 2 = 16{,}777{,}214$ hostids.
  $x_A = 127 \cdot 16{,}777{,}214 = 2{,}130{,}706{,}178$.
- To find, x_B: $2^{14} = 16{,}384$ netids. $2^{16} - 2 = 16{,}534$ hostids.
  $x_B = 16{,}384 \cdot 16{,}534 = 1{,}073{,}709{,}056$.
- To find, x_C: $2^{21} = 2{,}097{,}152$ netids. $2^8 - 2 = 254$ hostids.
  $x_C = 2{,}097{,}152 \cdot 254 = 532{,}676{,}608$.
- Hence, the total number of available IPv4 addresses is $x = x_A + x_B + Xc$
  $= 2{,}130{,}706{,}178 + 1{,}073{,}709{,}056 + 532{,}676{,}608 = 3{,}737{,}091{,}842$.

6.1.4 The Subtraction Rule (Inclusion–Exclusion for Two Sets)

Subtraction Rule: If a task can be done either in one of n_1 ways or in one of n_2 ways, then the total number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways.

Also known as, the *principle of inclusion-exclusion*:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

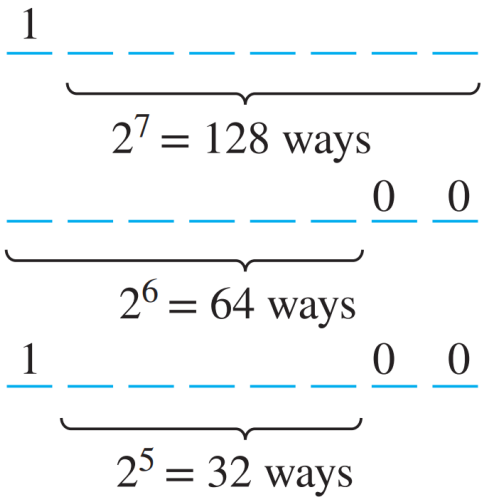
Counting Bit Strings

Example: How many bit strings of length eight either start with a 1 bit or end with the two bits 00?

Solution: Use the subtraction rule.

- Number of bit strings of length eight that start with a 1 bit: $2^7 = 128$
- Number of bit strings of length eight that end with bits 00: $2^6 = 64$
- Number of bit strings of length eight that start with a 1 bit and end with bits 00 : $2^5 = 32$

Hence, the number is $128 + 64 - 32 = 160$.



6.1.5 The Division Rule

Example: Suppose that there are exactly 572 legs in a farmer's pasture . How many cows are there is this pasture, assuming that each cow has four legs and that there are no other animals present?

572/4 = 143 cows.

Division Rule: There are n/d ways to do a task if it can be done using a procedure that can be carried out in n ways, and for every way w, exactly d of the n ways correspond to way w.

Restated in terms of sets: If the finite set A is the union of n pairwise disjoint subsets each with d elements, then $n = |A|/d$.

In terms of functions: If f is a function from A to B, where both are finite sets, and for <u>every value $y \in B$</u> there are <u>exactly d values $x \in A$</u> such that $f(x) = y$, then $|B| = |A|/d$.

The Division Rule

Example: How many ways are there to seat four people around a circular table, where two seating's are considered the same when each person has the same left and right neighbor?

Solution:

Number the seats around the table from 1 to 4 proceeding clockwise.

There are four ways to select the person for seat 1, three for seat 2, two for seat 3, and one way for seat 4.

Thus there are $4! = 24$ ways to order the four people.

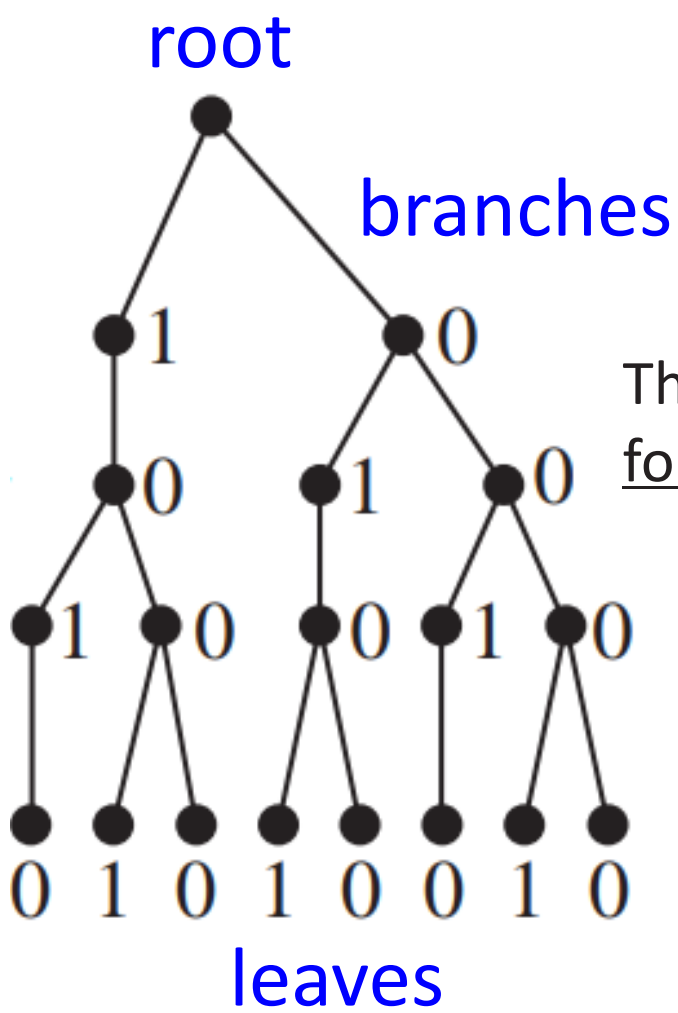
But since two seating's are the same when each person has the same left and right neighbor, for every choice for seat 1, we get the same seating.

Therefore, by the division rule, there are $24/4 = 6$ different seating arrangements.

6.1.6 Tree Diagrams

Counting problems can be solved using tree diagrams.

Tree

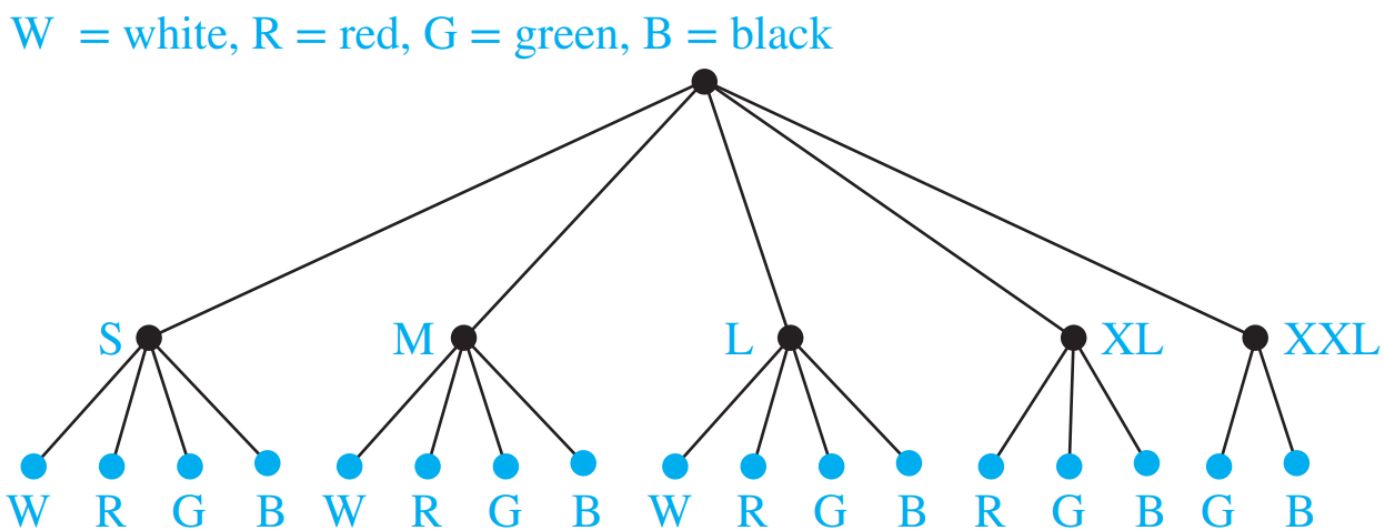


There are eight bit strings of length four without two consecutive 1s.

Tree Diagrams

Example: Suppose that "I Love Discrete Math" T-shirts come in five different sizes: S,M,L,XL, and XXL. Each size comes in four colors (white, red, green, and black), except XL, which comes only in red, green, and black, and XXL, which comes only in green and black. What is the minimum number of shirts that the campus book store needs to stock to have one of each size and color available?

Solution: Draw the tree diagram.



The store must stock 17 T-shirts.

Section Summary

The Product Rule

The Sum Rule

The Subtraction Rule

The Division Rule

Tree Diagrams

The Pigeonhole Principle

Section 6.2

6.2.1 The Pigeonhole Principle

If a flock of 20 pigeons roosts in a set of 19 pigeonholes, one of the pigeonholes must have more than 1 pigeon.

THEOREM 1 If k is a positive integer and $k + 1$ objects are placed into k boxes, then at least one box contains two or more objects.

Proof: We use a proof by contraposition. Suppose none of the k boxes has more than one object. Then the total number of objects would be at most k. This contradicts the statement that we have $k + 1$ objects.

The Pigeonhole Principle

Corollary 1: A function f from a set with $k + 1$ elements to a set with k elements is not one-to-one.

Proof: Use the pigeonhole principle.

- Create a box for each element y in the codomain of f .
- Put in the box for y all of the elements x from the domain such that $f(x) = y$.
- Because there are $k + 1$ elements and only k boxes, at least one box has two or more elements.

Hence, f can't be one-to-one.

Pigeonhole Principle

Example: Among any group of 367 people, there must be at least two with the same birthday.

Solution: Because there are only 366 possible birthdays.

Example : In any group of 27 English words, there must be at least two that begin with the same letter.

Solution: Because there are 26 letters in the English alphabet.

Pigeonhole Principle

Example : Show that for <u>every integer *n*</u> there is <u>a multiple of *n*</u> that has only 0s and 1s in its <u>decimal expansion</u>.

1=1/1 2=10/5 3=111/37 4=100/25 5=10/2 6=1110/185

7=1001/143 8=1000/125 9=111111111/12345679 10=10/1

Solution: Let *n* be a positive integer. Consider the $n + 1$ integers 1, 11, 111, …., 11…1 (where the last has $n + 1$ 1s).

There are *n* possible remainders when an integer is divided by *n*.

By the pigeonhole principle, when each of the $n + 1$ integers is divided by *n*, at least two must have the same remainder.

Subtract the smaller from the larger and the result is a multiple of *n* that has only 0s and 1s in its decimal expansion.

6.2.2 The Generalized Pigeonhole Principle

THEOREM 2: If N objects are placed into k boxes, then there is at least one box containing at least $\lceil N/k \rceil$ objects.

Proof: We use a proof by contraposition. Suppose that none of the boxes contains more than $\lceil N/k \rceil - 1$ objects. Then the total number of objects is at most

$$k\left(\left\lceil \frac{N}{k} \right\rceil - 1\right) < k\left(\left(\frac{N}{k} + 1\right) - 1\right) = N,$$

where the inequality $\lceil N/k \rceil < N/k + 1$ has been used.

This is a contradiction because there are a total of n objects.

Example: Among 100 people there are at least $\lceil 100/12 \rceil = 9$ who were born in the same month.

Poker

Example:(a) How many cards must be selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit are chosen?

The smallest integer N such that $\lceil N/4 \rceil \geq 3$ (Four suit)is $N = 2 \cdot 4 + 1 = 9$.

(b) How many must be selected to guarantee that at least three hearts are selected?

A deck contains 13 hearts and 39 cards which are not hearts.

So, when we select 42 cards, we must have at least three hearts.

Four suit

6.2.3 Some Elegant Applications of the Pigeonhole Principle

In many interesting applications of the pigeonhole principle, the objects to be placed in boxes must be chosen in a clever way.

Some Elegant Applications of the Pigeonhole Principle

Example: During a month with 30 days, a baseball team plays at least one game a day, but no more than 45 games. Show that there must be a period of some number of consecutive days during which the team must play exactly 14 games.

Solution: Let a_j be the number of games played on or before the jth day of the month.

Then $a_1, a_2, \ldots, a_{30}$ is an increasing sequence of distinct integers, with $1 \leq a_j \leq 45$.

More over, $a_1 + 14, a_2 + 14, \ldots, a_{30} + 14$ is also an increasing sequence of distinct positive integers, with $15 \leq aj + 14 \leq 59$.

The 60 positive integers $a_1, a_2, \ldots, a_{30}, a_1 + 14, a_2 + 14, \ldots, a_{30} + 14$ are all less than or equal to 59.

Hence, by the pigeonhole principle two of these integers are equal.

Because the integers a_j, $j = 1, 2, \ldots, 30$ are all distinct and the integers $a_j + 14$, $j = 1, 2, \ldots, 30$ are all distinct, there must be indices i and j with $a_i = a_j + 14$.

This means that exactly 14 games were played from day $j + 1$ to day i.

Strictly Increasing and Strictly Decreasing

A clever application of the pigeonhole principle shows the existence of an increasing or a decreasing subsequence of a certain length in a sequence of distinct integers. We review some definitions before this application is presented. Suppose that $a_1, a_2, \ldots, a_N$ is a sequence of real numbers. A subsequence of this sequence is a sequence of the form $a_{i1}, a_{i2}, \ldots, a_{im}$, where $1 \leq i_1 < i_2 < \cdots < i_m \leq N$.

Hence, a subsequence is a sequence obtained from the original sequence by including some of the terms of the original sequence in their original order, and perhaps not including other terms.

A sequence is called strictly increasing if each term is larger than the one that precedes it, and it is called strictly decreasing if each term is smaller than the one that precedes it.

Strictly Increasing and Strictly Decreasing

THEOREM 3: Every sequence of $n^2 + 1$ distinct real numbers <u>contains a</u> subsequence of length $n + 1$ that is either strictly increasing or strictly decreasing.

Example: The sequence 8, 11, 9, 1, 4, 6, 12, 10, 5, 7 contains 10 terms.

Note that $10 = 3^2 + 1$.

There are four strictly increasing subsequences of length four, namely,

1, 4, 6, 12; 1, 4, 6, 7; 1, 4, 6, 10; 1, 4, 5, 7.

There is also a strictly decreasing subsequence of length four, namely,

11, 9, 6, 5.

Section Summary

The Pigeonhole Principle

The Generalized Pigeonhole Principle

Permutations and Combinations

Section 6.3

Arrange

Permutations and Combinations

$$A_n^m = n(n-1)(n-2)\cdots(n-m+1)$$

$$A_n^m = \frac{n!}{(n-m)!}$$

$$C_n^m = \frac{A_n^m}{m!} = \frac{n(n-1)\cdots(n-m+1)}{m!}$$

$$C_n^m = \frac{n!}{m!(n-m)!}$$

$$C_{n+1}^m = C_n^m + C_n^{m-1}$$

$$C_n^m = C_n^{n-m}$$

6.3.1 Introduction

Many counting problems can be solved by finding the number of ways to arrange a specified number of distinct elements of a set of a particular size, where the order of these elements matters.

- ✓ How many ways can we select three students from a group of five students to stand in line for a picture?

Many other counting problems can be solved by finding the number of ways to select a particular number of elements from a set of a particular size, where the order of the elements selected does not matter.

- ✓ How many different committees of three students can be formed from a group of four students?

6.3.2 Permutations

A permutation of a set of distinct objects is an ordered arrangement of these objects.

An ordered arrangement of r elements of a set is called an r-permutation.

Example: Let S = {1,2,3}.

- The ordered arrangement 3,1,2 is a <u>permutation of S</u>.
- The ordered arrangement 3,2 is a <u>2-permutation of S</u>.

The number of r-permutations of a set with *n* elements is denoted by $P(n,r)$.

- The 2-permutations of $S = \{1,2,3\}$ are <u>1,2; 1,3; 2,1; 2,3; 3,1; and 3,2</u>. Hence, $P(3,2) = 3 \cdot 2 = 6$.

A Formula for the Number of Permutations

Theorem 1: If n is a positive integer and r is an integer with $1 \leq r \leq n$, then there are

$$P(n, r) = n(n-1)(n-2) \cdots (n-r+1)$$

r-permutations of a set with n distinct elements.

Proof: Use the product rule. The first element can be chosen in n ways. The second in $n - 1$ ways, and so on until there are $(n - (r - 1))$ ways to choose the last element.

Note that $P(n,0) = 1$, since there is only one way to order zero elements.

Corollary 1: If n and r are integers with $1 \leq r \leq n$, then

$$P(n,r) = \frac{n!}{(n-r)!}$$

Solving Counting Problems by Counting Permutations

Example: How many ways are there to select a first-prize winner, a second prize winner, and a third-prize winner from 100 different people who have entered a contest?

Solution: $P(100,3) = 100 \cdot 99 \cdot 98 = 970{,}200$

Example: How many permutations of the letters *ABCDEFGH* contain the string *ABC* ?

Solution: We solve this problem by counting the permutations of six objects, *ABC*, *D*, *E*, *F*, *G*, and *H*.

$$6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720$$

Solving Counting Problems by Counting Permutations

Example: Suppose that a saleswoman has to visit eight different cities. She must begin her trip in a specified city, but she can visit the other seven cities in any order she wishes. How many possible orders can the saleswoman use when visiting these cities?

Solution: The first city is chosen, and the rest are ordered arbitrarily. Hence the orders are:

$$7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5040$$

If she wants to find the tour with the shortest path that visits all the cities, she must consider 5040 paths!

6.3.3 Combinations

We now turn our attention to counting unordered selections of objects.

An *r-combination* of elements of a set is an unordered selection of *r* elements from the set.

Thus, an *r*-combination is simply a subset of the set with *r* elements.

The number of *r*-combinations of a set with n distinct elements is denoted by $C(n, r)$.

The notation $\binom{n}{r}$ is also used and is called a *binomial coefficient*.

Example: Let *S* be the set {*a*, *b*, *c*, *d*}.

Then {*a*, *c*, *d*} is a 3-combination from S. It is the same as {*d*, *c*, *a*}.

$C(4,2) = 6$ because the 2-combinations of {*a*, *b*, *c*, *d*} are the six subsets {*a*, *b*}, {*a*, *c*}, {*a*, *d*}, {*b*, *c*}, {*b*, *d*}, and {*c*, *d*}.

Combinations

Theorem 2: The number of r-combinations of a set with n elements, where $n \geq r \geq 0$, equals

$$C(n,r) = \frac{n!}{(n-r)!r!}.$$

Proof: By the product rule $P(n, r) = C(n,r) \cdot P(r,r)$.

$$C(n, r) = \frac{P(n, r)}{P(r, r)} = \frac{n!/(n-r)!}{r!/(r-r)!} = \frac{n!}{r!\,(n-r)!}.$$

$$C(n, r) = \frac{n!}{r!\,(n-r)!} = \frac{n(n-1)\cdots(n-r+1)}{r!}.$$

Combinations

Example: How many poker hands of five cards can be dealt from a standard deck of 52 cards? Also, how many ways are there to select 47 cards from a deck of 52 cards?

Solution: Since the order in which the cards are dealt does not matter, the number of five card hands is:

$$C(52, 5) = \frac{52!}{5!47!} = \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}.$$

$$= 26 \cdot 17 \cdot 10 \cdot 49 \cdot 12 = 2{,}598{,}960.$$

The different ways to select 47 cards from 52 is

$$C(52, 47) = \frac{52!}{47!5!} = C(52, 5) = 2{,}598{,}960.$$

Combinations

Corollary 2: Let n and r be nonnegative integers with $r \leq n$. Then

$$C(n, r) = C(n, n - r).$$

Proof: From Theorem 2, it follows that

$$\boxed{C(n, r) = \frac{n!}{r!\,(n-r)!}}$$

and

$$\boxed{C(n, n-r) = \frac{n!}{(n-r)!\,[n-(n-r)]!} = \frac{n!}{(n-r)!\,r!}.}$$

Hence, $C(n, r) = C(n, n - r)$.

Combinations

Example: How many ways are there to select five players from a 10-member tennis team to make a trip to a match at another school.

$$C(10,5) = \frac{10!}{5!5!} = 252.$$

Example: A group of 30 people have been trained as astronauts to go on the first mission to Mars. How many ways are there to select a crew of six people to go on this mission?

$$C(30,6) = \frac{30!}{6!24!} = \frac{30 \cdot 29 \cdot 28 \cdot 27 \cdot 26 \cdot 25}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 593,775$$

Combinations

Example: How many bit strings of length n contain exactly r 1s?

$$C(n, r)$$

Example: Suppose that there are 9 faculty members in the mathematics department and 11 in the computer science department. How many ways are there to select a committee to develop a discrete mathematics course at a school if the committee is to consist of three faculty members from the mathematics department and four from the computer science department?

$$C(9, 3) \cdot C(11, 4) = \frac{9!}{3!6!} \cdot \frac{11!}{4!7!} = 84 \cdot 330 = 27{,}720$$

Section Summary

Permutations
Combinations
Combinatorial Proofs

Binomial Coefficients and Identities

Section 6.4

6.4.1 The Binomial Theorem

The number of *r*-combinations from a set with *n* elements is often denoted by $\binom{n}{r}$.

This number is also called a binomial coefficient because these numbers occur as coefficients in the expansion of powers of binomial expressions such as $(a + b)^n$.

$$\begin{aligned}(x+y)^3 &= (x+y)(x+y)(x+y) = (xx+xy+yx+yy)(x+y)\\ &= xxx+xxy+xyx+xyy+yxx+yxy+yyx+yyy\\ &= x^3+3x^2y+3xy^2+y^3.\end{aligned}$$

Binomial Theorem

THEOREM 1: Let x and y be variables, and n a nonnegative integer. Then:

$$(x+y)^n = \sum_{j=0}^{n} \binom{n}{j} x^{n-j} y^j = \binom{n}{0} x^n + \binom{n}{1} x^{n-1} y + \cdots + \binom{n}{n-1} x y^{n-1} + \binom{n}{n} y^n.$$

Proof: We use combinatorial reasoning .

The terms in the expansion of $(x + y)^n$ are of the form $x^{n-j}y^j$ for j = 0,1,2,…,n.

To form the term $x^{n-j}y^j$, it is necessary to choose $n-j$ x from the n sums.

Therefore, the coefficient of $x^{n-j}y^j$ is $\binom{n}{n-j}$ which equals $\binom{n}{j}$

Using the Binomial Theorem

Example: What is the expansion of $(x + y)^4$?

$$(x+y)^4 = \sum_{j=0}^{4} \binom{4}{j} x^{4-j} y^j$$

$$= \binom{4}{0} x^4 + \binom{4}{1} x^3 y + \binom{4}{2} x^2 y^2 + \binom{4}{3} x y^3 + \binom{4}{4} y^4$$

$$= x^4 + 4x^3 y + 6x^2 y^2 + 4xy^3 + y^4.$$

Example: What is the coefficient of $x^{12}y^{13}$ in the expansion of $(x + y)^{25}$?

$$\binom{25}{13} = \frac{25!}{13!\,12!} = 5{,}200{,}300.$$

Example: What is the coefficient of $x^{12}y^{13}$ in the expansion of $(2x - 3y)^{25}$?

$$(2x + (-3y))^{25} = \sum_{j=0}^{25} \binom{25}{j} (2x)^{25-j} (-3y)^j.$$

$$\binom{25}{13} 2^{12} (-3)^{13} = -\frac{25!}{13!\,12!} 2^{12} 3^{13}.$$

A Useful Identity

COROLLARY 1: Let *n* be a nonnegative integer. Then $\sum_{k=0}^{n} \binom{n}{k} = 2^n$.

$$2^n = (1+1)^n = \sum_{k=0}^{n} \binom{n}{k} 1^k 1^{n-k} = \sum_{k=0}^{n} \binom{n}{k}.$$

COROLLARY 2: Let n be a positive integer. Then $\sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0$.

$$0 = 0^n = ((-1)+1)^n = \sum_{k=0}^{n} \binom{n}{k} (-1)^k 1^{n-k} = \sum_{k=0}^{n} \binom{n}{k} (-1)^k.$$

COROLLARY 3: Let n be a nonnegative integer. Then $\sum_{k=0}^{n} 2^k \binom{n}{k} = 3^n$.

$$(1+2)^n = \sum_{k=0}^{n} \binom{n}{k} 1^{n-k} 2^k = \sum_{k=0}^{n} \binom{n}{k} 2^k.$$

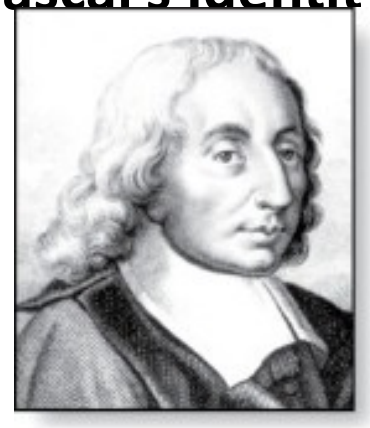
6.4.2 Pascal's Identity and Triangle

The binomial coefficients satisfy many different identities.

We introduce one of the most important of these now (Pascal's Identity).

Pascal's Identity: If n and k are integers with $n \geq k \geq 0$, then

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$$



Blaise Pascal
(1623-1662)

Pascal's Identity

Pascal's Identity: If n and k are integers with $n \geq k \geq 0$, then $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$

Proof (*combinatorial*): Let T be a set where $|T| = n + 1$, $a \in T$, and $S = T - \{a\}$.

There are $\binom{n+1}{k}$ subsets of T containing k elements.

Each of these subsets either:

- contains a with $k - 1$ other elements, or
- contains k elements of S and not a.

There are

- $\binom{n}{k-1}$ subsets of k elements that contain a, since there are $\binom{n}{k-1}$ subsets of k − 1 elements of S,
- $\binom{n}{k}$ subsets of k elements of T that do not contain a, because there are $\binom{n}{k}$ subsets of k elements of S.

Hence, $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$.

Yang Hui triangle

Pascal's Triangle

The *n*th row in the triangle consists of the binomial coefficients $\binom{n}{k}$, k = 0,1,….,*n*.

$$\binom{0}{0}$$

$$\binom{1}{0}\ \binom{1}{1}$$

$$\binom{2}{0}\ \binom{2}{1}\ \binom{2}{2}$$

$$\binom{3}{0}\ \binom{3}{1}\ \binom{3}{2}\ \binom{3}{3}$$

$$\binom{4}{0}\ \binom{4}{1}\ \binom{4}{2}\ \binom{4}{3}\ \binom{4}{4}$$

$$\binom{5}{0}\ \binom{5}{1}\ \binom{5}{2}\ \binom{5}{3}\ \binom{5}{4}\ \binom{5}{5}$$

$$\binom{6}{0}\ \binom{6}{1}\ \binom{6}{2}\ \binom{6}{3}\ \binom{6}{4}\ \binom{6}{5}\ \binom{6}{6}$$

$$\binom{7}{0}\ \binom{7}{1}\ \binom{7}{2}\ \binom{7}{3}\ \binom{7}{4}\ \binom{7}{5}\ \binom{7}{6}\ \binom{7}{7}$$

$$\binom{8}{0}\ \binom{8}{1}\ \binom{8}{2}\ \binom{8}{3}\ \binom{8}{4}\ \binom{8}{5}\ \binom{8}{6}\ \binom{8}{7}\ \binom{8}{8}$$

$$\cdots$$

By Pascal's identity:

$$\binom{6}{4} + \binom{6}{5} = \binom{7}{5}$$

1

1 1

1 2 1

1 3 3 1

1 4 6 4 1

1 5 10 10 5 1

1 6 15 20 15 6 1

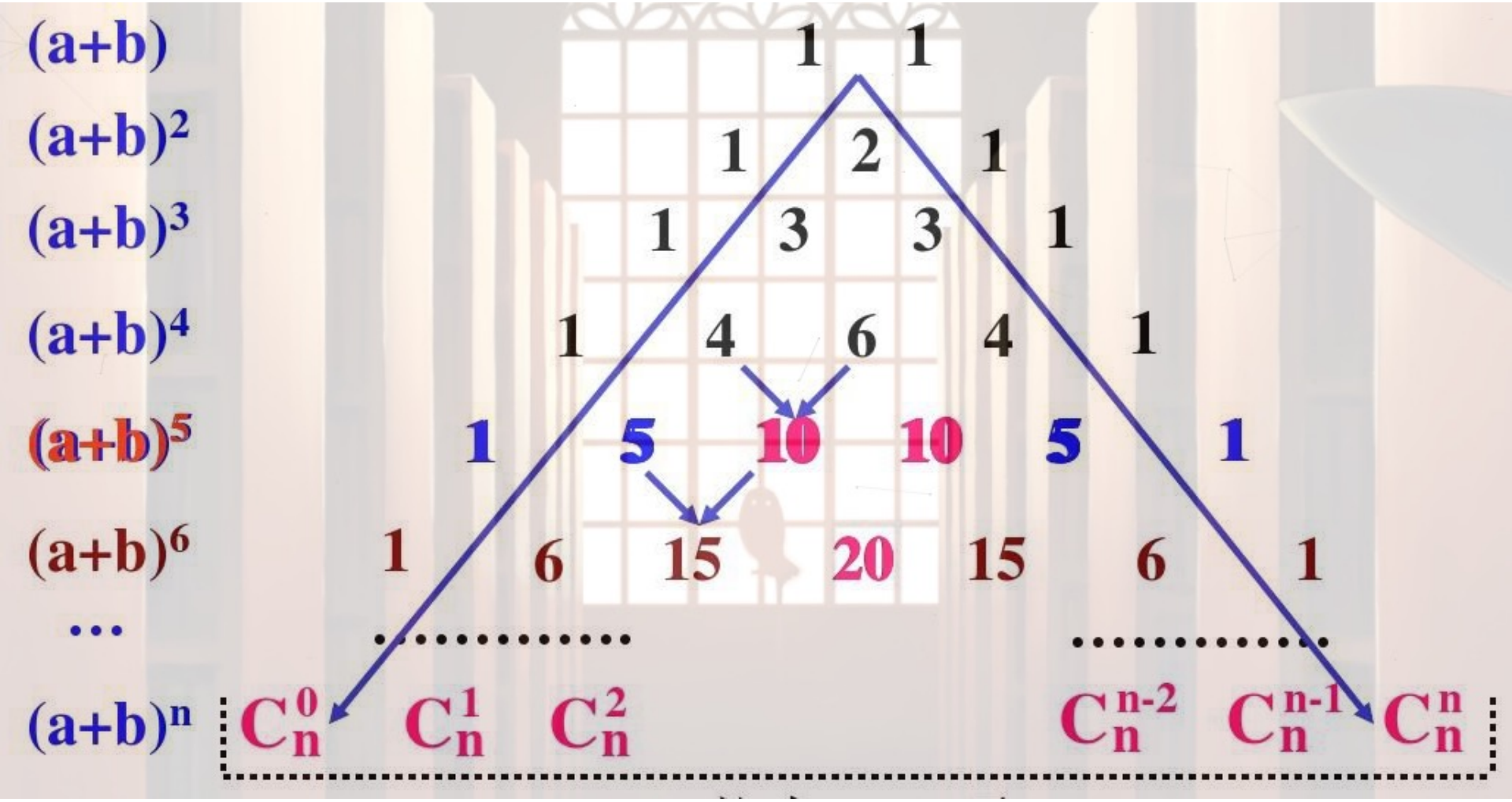
1 7 21 35 35 21 7 1

1 8 28 56 70 56 28 8 1

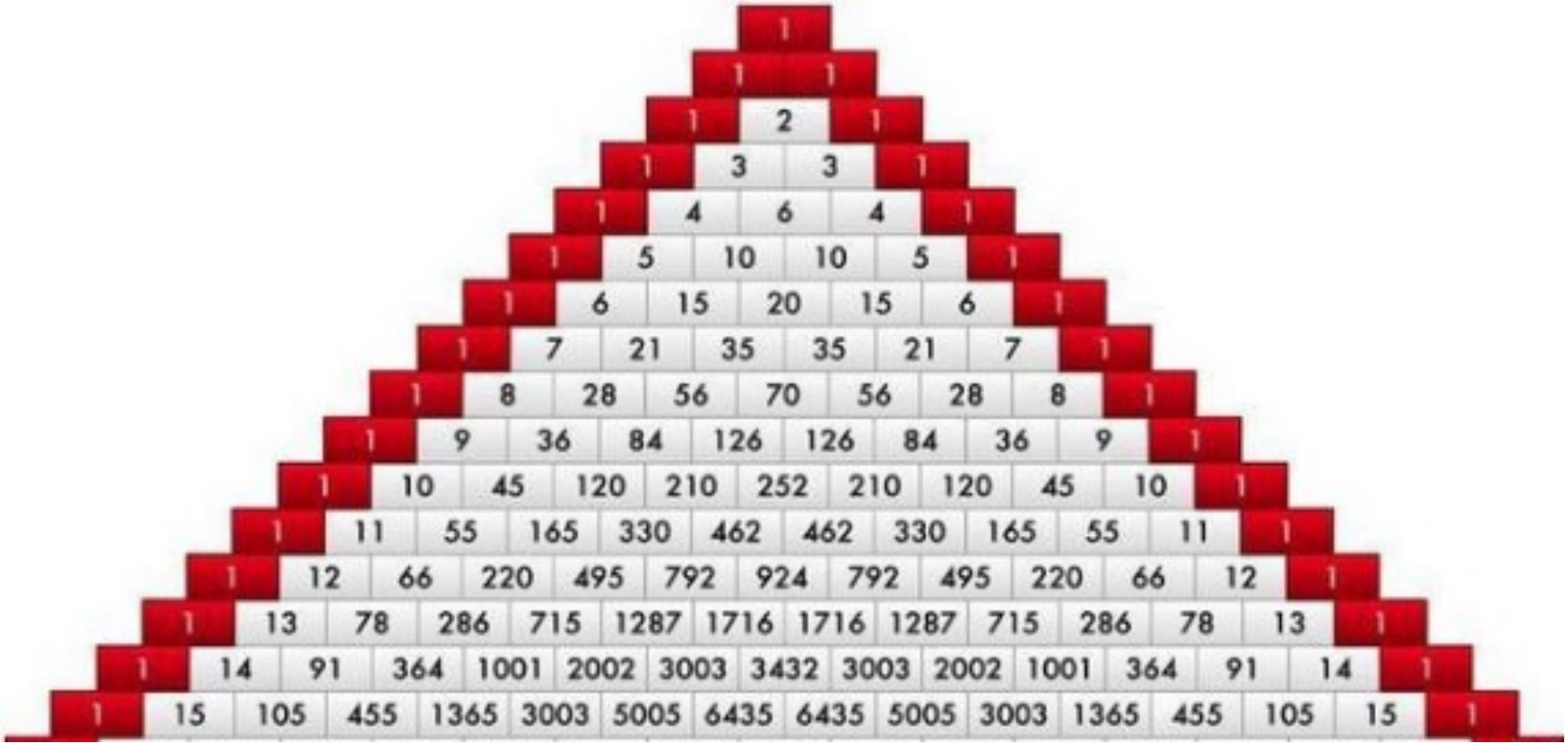
. . .

By Pascal's identity, adding two adjacent binomial coefficients results is the binomial coefficient in the next row between these two coefficients.

Yang Hui Triangle



Yang Hui Triangle



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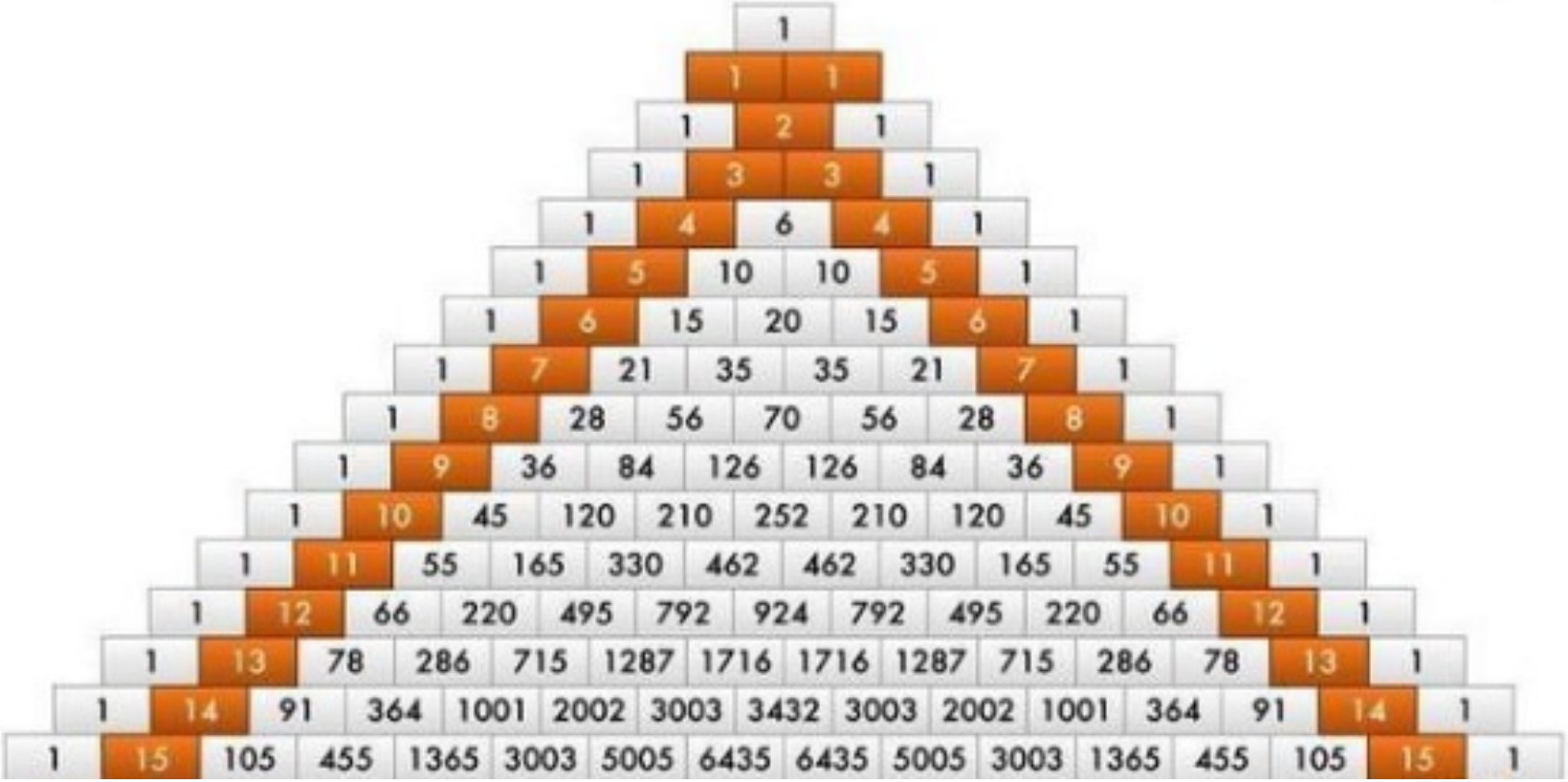

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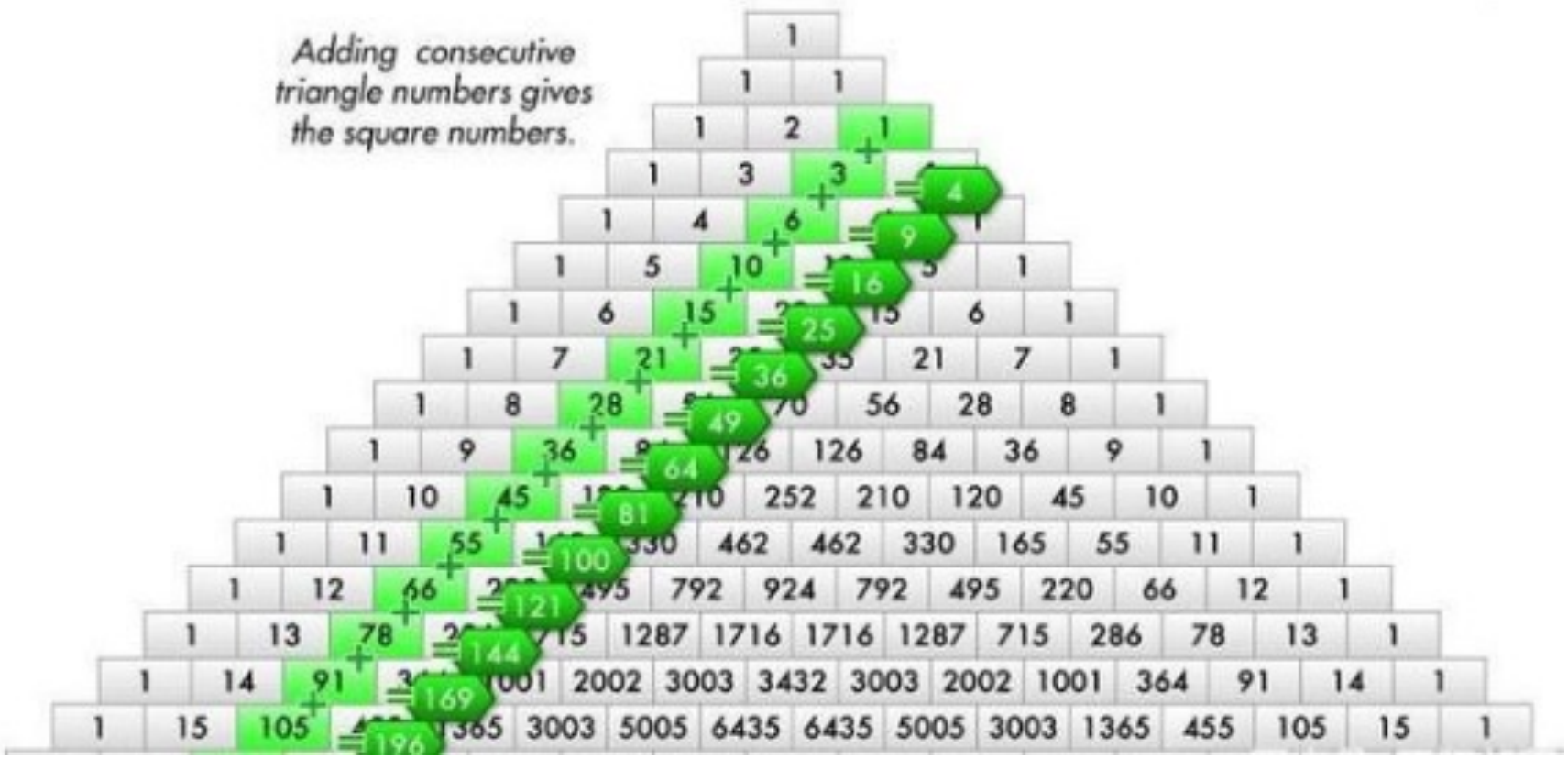

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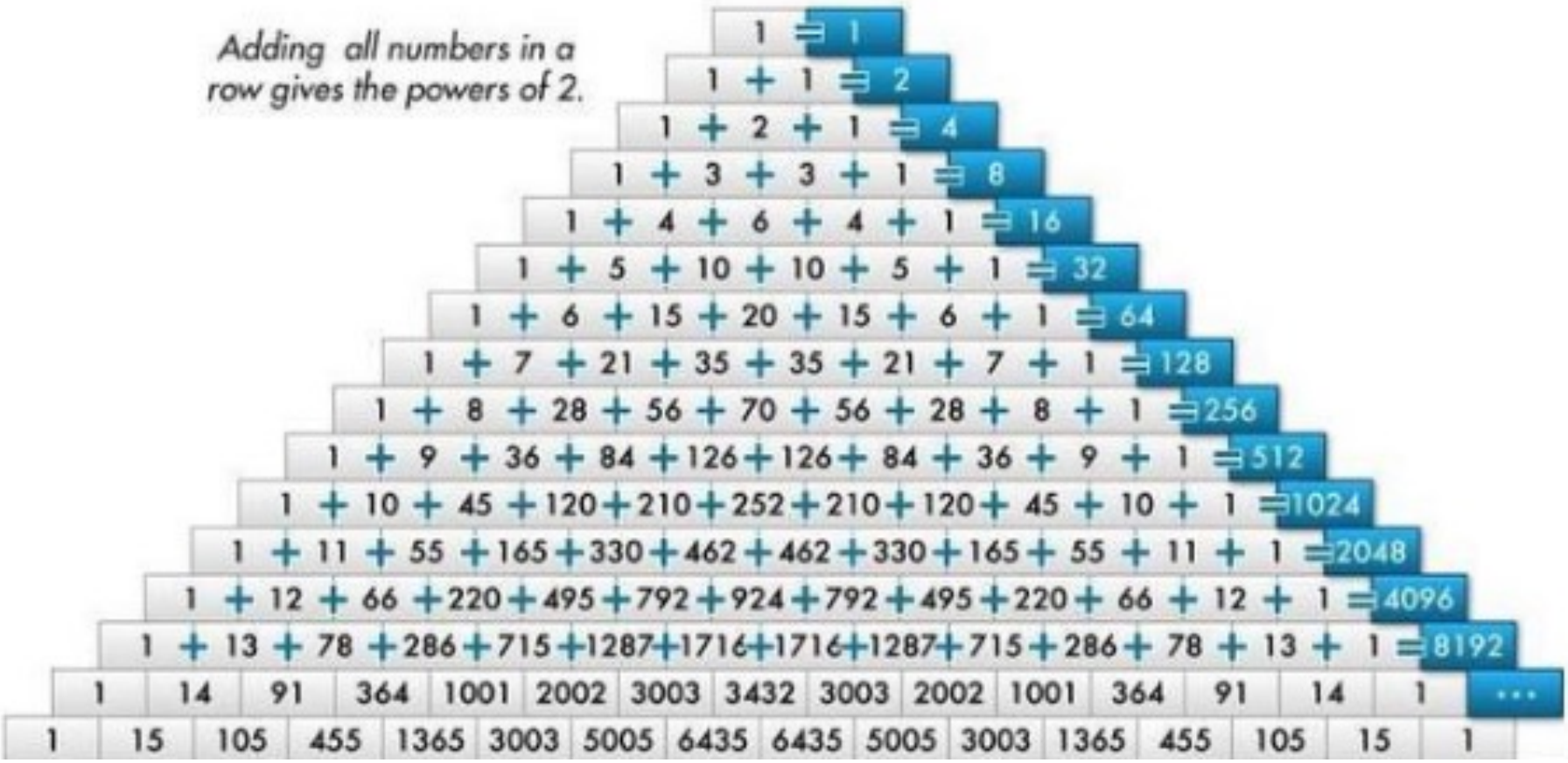

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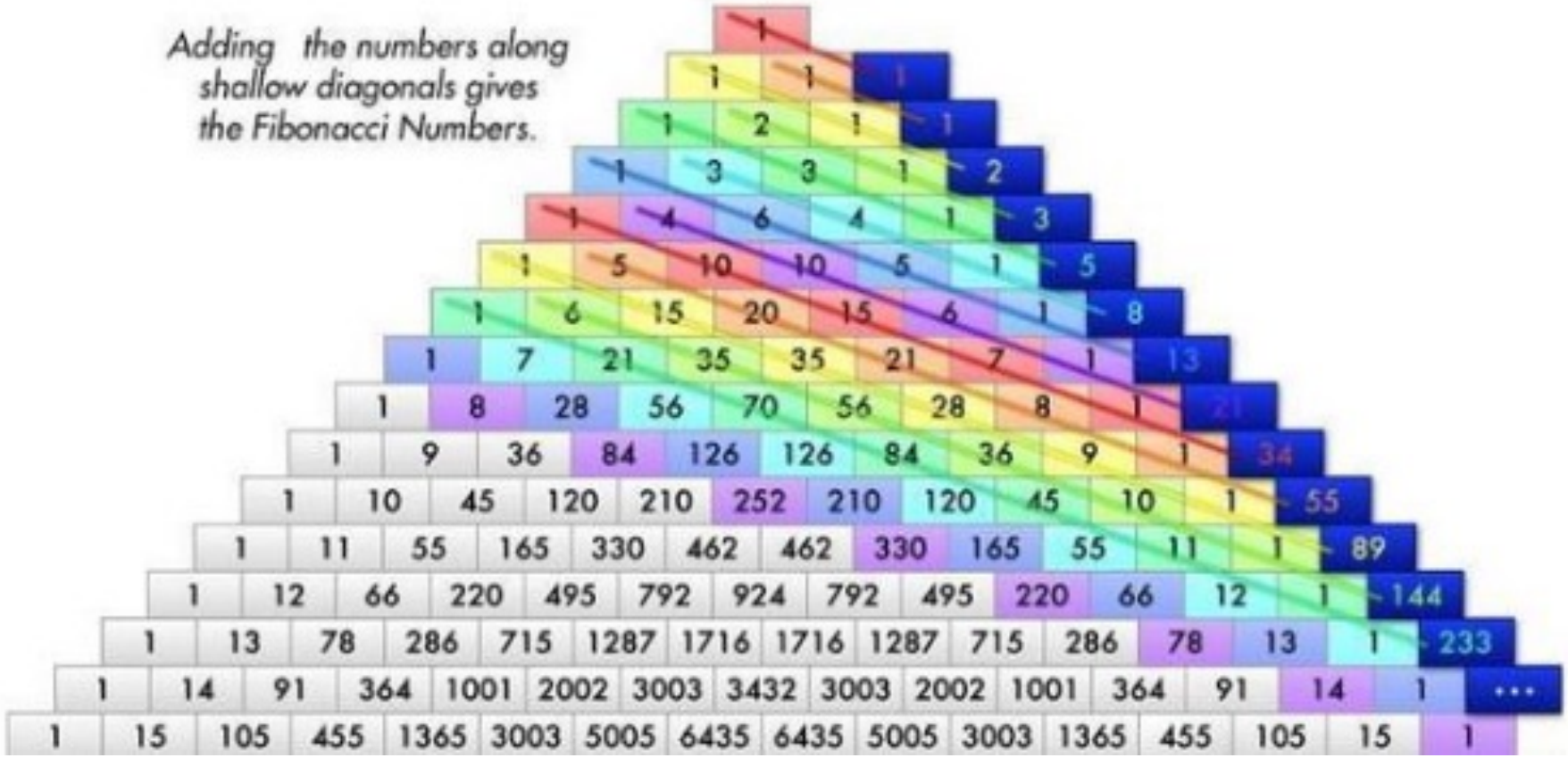
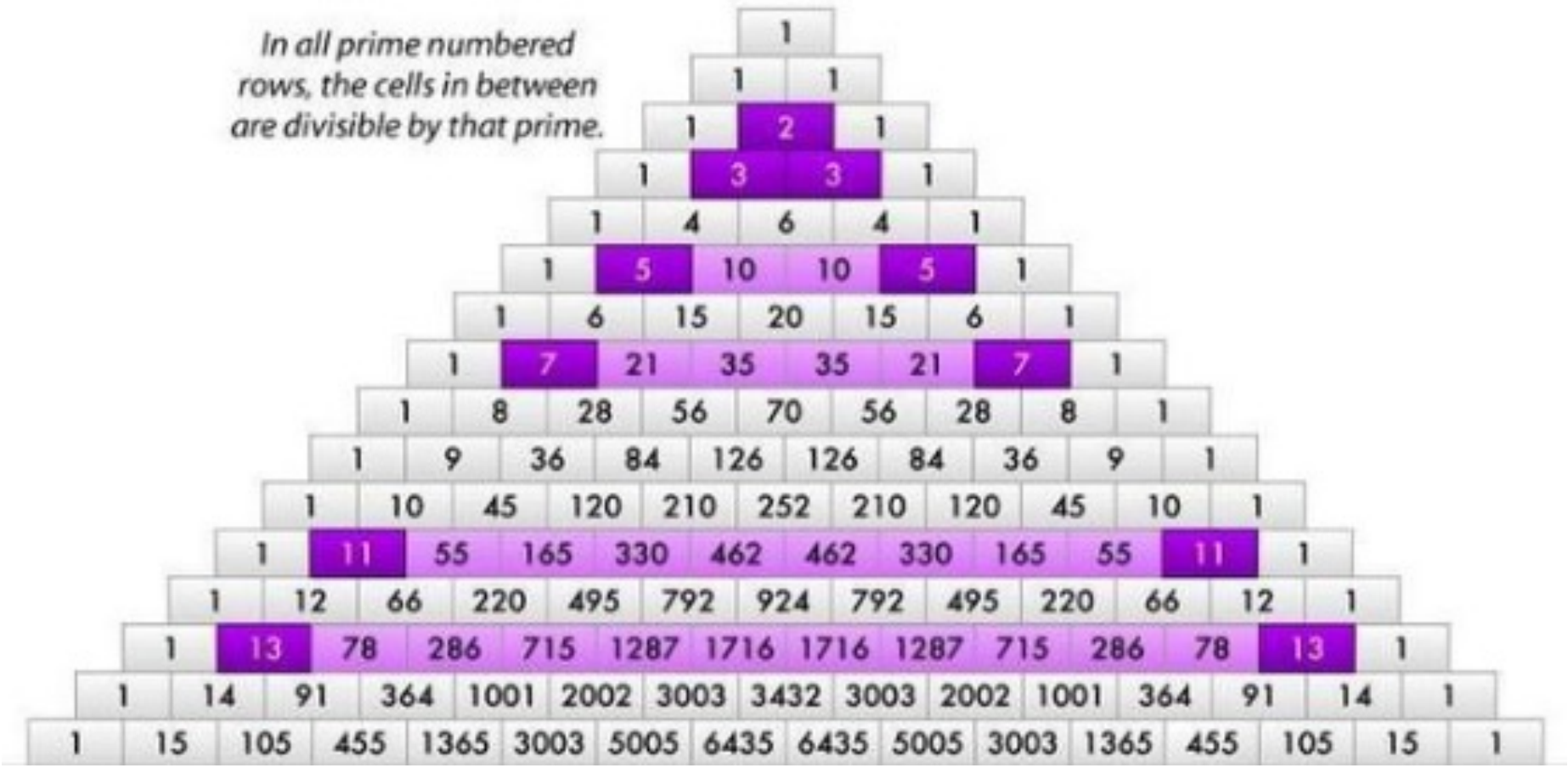


Yang Hui Triangle



6.4.3 Other Identities Involving Binomial Coefficients

THEOREM 3: VANDERMONDE'S IDENTITY Let *m*, *n*, and *r* be nonnegative integers with *r* not exceeding either *m* or *n*. Then

$$\binom{m+n}{r} = \sum_{k=0}^{r} \binom{m}{r-k}\binom{n}{k}.$$

COROLLARY 4: Let n be nonnegative integers. Then

$$\binom{2n}{n} = \sum_{k=0}^{n} \binom{n}{k}^2.$$

THEOREM 4: Let n and r be nonnegative integers with r ≤ n. Then

$$\binom{n+1}{r+1} = \sum_{j=r}^{n} \binom{j}{r}.$$

Section Summary

The Binomial Theorem

Pascal's Identity and Triangle

Other Identities Involving Binomial Coefficients

Generalized Permutations and Combinations

Section 6.5

6.5.1 Introduction

In many counting problems, elements may be used repeatedly.

- ✓ A letter or digit may be used more than once on a license plate.

Also, some counting problems involve indistinguishable elements.

- ✓ To count the number of ways the letters of the word SUCCESS can be rearranged, the placement of identical letters must be considered

Moreover, problems involving counting the ways distinguishable elements can be placed in boxes.

- ✓ The number of different ways poker hands can be dealt to four players.

6.5.2 Permutations with Repetition

Example: How many strings of length r can be formed from the uppercase letters of the English alphabet?

Solution: The number of such strings is 26^r, which is the number of r-permutations of a set with 26 elements.

Theorem 1: The number of r-permutations of a set of n objects with repetition allowed is n^r.

Proof:

There are n ways to select an element of the set for each of the r positions in the r-permutation when repetition is allowed.

Hence, by the product rule there are n^r r-permutations with repetition.

6.5.3 Combinations with Repetition

How many ways are there to select four pieces of fruit from a bowl containing apples, oranges, and pears if the order in which the pieces are selected does not matter, only the type of fruit and not the individual piece matters, and there are four pieces in the bowl?

Solution: The solution is the number of 4-combinations with repetition allowed from a three-element set, {*apple*, *orange*, *pear*}.

To solve this problem we list all the ways possible to select the fruit.

4 apples

4 oranges

4 pears

3 apples, 1 orange

3 apples, 1 pear

3 oranges, 1 apple

3 oranges, 1 pear

3 pears, 1 apple

3 pears, 1 orange

2 apples, 2 oranges

2 apples, 2 pears

2 oranges, 2 pears

2 apples, 1 orange, 1 pear

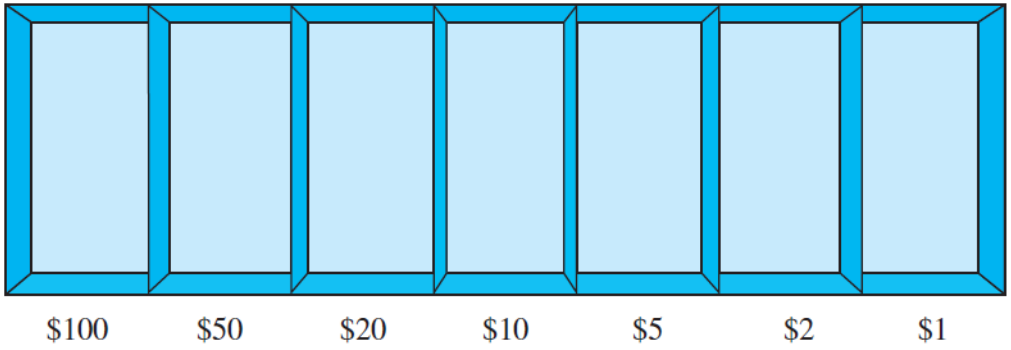
2 oranges, 1 apple, 1 pear

2 pears, 1 apple, 1 orange

Combinations with Repetition

Example: How many ways are there to select five bills from a box containing at least five of each of the following denominations: $1, $2, $5, $10, $20, $50, and $100?

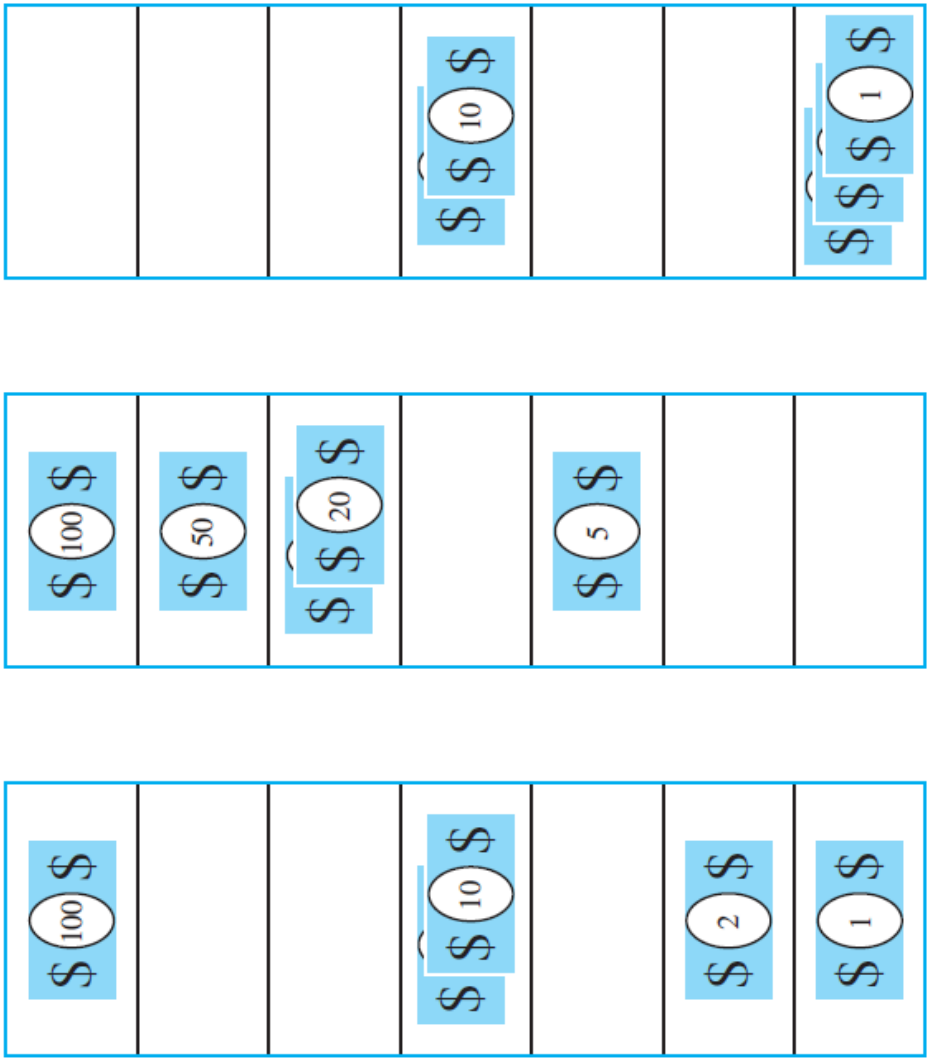
Solution:



Here are seven different types of bills can be selected as many as five times, this problem involves counting 5-combinations with repetition allowed from a set with seven elements.

Listing all possibilities would be tedious, because there are a large number of solutions.

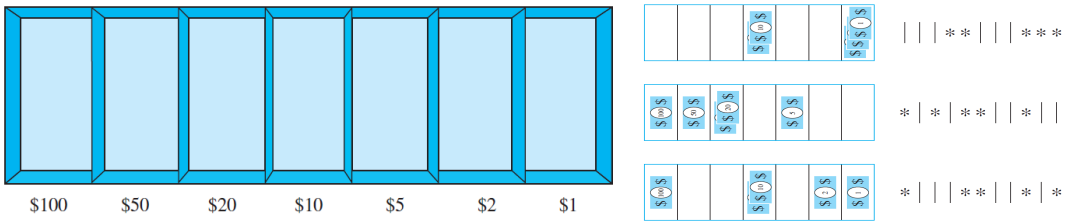
Combinations with Repetition



*				*	*			*		*
*		*		*	*			*		
*				*	*			*		*

Combinations with Repetition

Example: How many ways are there to select five bills from a box containing at least five of each of the following denominations: $1, $2, $5, $10, $20, $50, and $100?



Solution:

The number of ways to select five bills corresponds to the number of ways to arrange six bars and five stars in a row with a total of 11 positions.

Consequently, the number of ways to select the five bills is the number of ways to select the positions of the five stars from the 11 positions.

This corresponds to the number of unordered selections of 5 objects from a set of 11 objects, which can be done in $C(11, 5)$ ways.

Consequently, there are $C(11, 5) = 11! = 462$ ways to choose five bills from the cash box with seven types of bills.

Combinations with Repetition

Theorem 2: There are $C(n + r - 1, r) = C(n + r - 1, n - 1)$ r-combinations from a set with n elements when repetition of elements is allowed.

Proof:

Each *r*-combination of a set with *n* elements with repetition allowed can be represented by a list of $n - 1$ bars and r stars.

The bars mark the *n* cells containing a star for each time the *i*th element of the set occurs in the combination.

The number of such lists is $C(n + r - 1, r)$, because each list is a choice of the *r* positions to place the stars, from the total of $n + r - 1$ positions to place the stars and the bars.

This is also equal to $C(n + r - 1, n - 1)$, which is the number of ways to place the $n - 1$ bars.

Combinations with Repetition

Example: Suppose that a cookie shop has four different kinds of cookies. How many different ways can six cookies be chosen? Assume that only the type of cookie, and not the individual cookies or the order in which they are chosen, matters.

Solution:

The number of ways to choose six cookies is the number of 6-combinations of a set with four elements.

By Theorem 2: $C(9,6) = C(9,3) = \frac{9 \cdot 8 \cdot 7}{1 \cdot 2 \cdot 3} = 84$ is the number of ways to choose six cookies from the four kinds.

Combinations with Repetition

Example: How many solutions does the equation $x_1 + x_2 + x_3 = 11$ have, where x_1 , x_2 and x_3 are nonnegative integers?

Solution:

Each solution corresponds to a way to select 11 items from a set with three elements; x_1 elements of type one, x_2 of type two, and x_3 of type three.

By Theorem 2 it follows that there are $C(3 + 11 - 1,11) = C(13,11) = C(13,2) = \frac{13 \cdot 2}{1 \cdot 2} = 78$ solutions.

Combinations with Repetition

Example: What is the value of k after the following pseudocode has been executed?

```
k := 0
for i_1 := 1 to n
    for i_2 := 1 to i_1
        .
        .
        .
        for i_m := 1 to i_{m-1}
            k := k + 1
```

$k = C(n + m - 1, m)$

Summarizing the Formulas for Counting Permutations and Combinations with and without Repetition

Type	***Repetition Allowed?***	***Formula***
r-permutations	No $P(n, r)$	$\frac{n!}{(n-r)!}$
r-combinations	No $C(n, r)$	$\frac{n!}{r!(n-r)!}$
r-permutations	Yes	n^r
r-combinations	Yes $C(n+r-1, r)$	$\frac{(n+r-1)!}{r!(n-1)!}$

6.5.4 Permutations with Indistinguishable Objects

Example: How many different strings can be made by reordering the letters of the word *SUCCESS*.

Solution: There are seven possible positions for the three Ss, two Cs, one U, and one E.

- The three Ss can be placed in $C(7,3)$ different ways, leaving four positions free.
- The two Cs can be placed in $C(4,2)$ different ways, leaving two positions free.
- The U can be placed in $C(2,1)$ different ways, leaving one position free.
- The E can be placed in $C(1,1)$ way.

By the product rule, the number of different strings is:

$$C(7,3)C(4,2)C(2,1)C(1,1) = \frac{7!}{3!4!} \cdot \frac{4!}{2!2!} \cdot \frac{2!}{1!1!} \cdot \frac{1!}{3!2!1!1!} = 420.$$

Permutations with Indistinguishable Objects

Theorem 3: The number of different permutations of n objects, where there are n_1 indistinguishable objects of type 1, n_2 indistinguishable objects of type 2, …., and n_k indistinguishable objects of type k, is:

$$\frac{n!}{n_1!\,n_2!\cdots n_k!}.$$

Proof: By the product rule the total number of permutations is:

$C(n, n_1)\, C(n - n_1, n_2) \cdots C(n - n_1 - n_2 - \cdots - n_k, n_k)$ since:

- The n_1 objects of type one can be placed in the n positions in $C(n, n_1)$ ways, leaving $n - n_1$ positions.
- Then the n_2 objects of type two can be placed in the $n - n_1$ positions in $C(n - n_1, n_2)$ ways, leaving $n - n_1 - n_2$ positions.
- Continue in this fashion, until n_k objects of type k are placed in $C(n - n_1 - n_2 - \cdots - n_k, n_k)$ ways.

The product can be manipulated into the desired result as follows:

$$\frac{n!}{n_1!\,(n-n_1)!}\frac{(n-n_1)!}{n_2!\,(n-n_1-n_2!)}\cdots\frac{(n-n_1-\cdots-n_{k-1})!}{n_k!\,0!}=\frac{n!}{n_1!\,n_2!\cdots n_k!}.$$

6.5.5 Distributing Objects into Boxes

Many counting problems can be solved by counting the ways objects can be placed in boxes.

- The objects may be either different from each other (*distinguishable*) or identical (*indistinguishable*).
- The boxes may be labeled (*distinguishable*) or unlabeled (*indistinguishable*).

✓ distinguishable objects and distinguishable boxes

✓ indistinguishable objects and distinguishable boxes

✓ distinguishable objects and indistinguishable boxes

✓ indistinguishable objects and indistinguishable boxes

Distinguishable Objects and Distinguishable Boxes

EXAMPLE: How many ways are there to distribute hands of 5 cards to each of four players from the standard deck of 52 cards?

Solution:

- ✓ The first player can be dealt 5 cards in C(52, 5) ways.
- ✓ The second player can be dealt 5 cards in C(47, 5) ways.
- ✓ The third player can be dealt 5 cards in C(42, 5) ways.
- ✓ Finally, the fourth player can be dealt 5 cards in C(37, 5) ways.

Hence, the total number of is C(52, 5) C(47, 5) C(42, 5) C(37,5)= $\frac{52!}{47!5!}\frac{47!}{42!5!}\frac{42!}{37!5!}\frac{37!}{32!5!}$.

THEOREM 4: The number of ways to distribute n distinguishable objects into k distinguishable boxes so that n_i objects are placed into box i, i = 1, 2,…, k, equals $\frac{n!}{n_1!n_2!\cdots n_k!}$.

Indistinguishable Objects and Distinguishable Boxes

EXAMPLE : How many ways are there to place 10 indistinguishable balls into eight distinguishable bins?

Solution:

The number of ways to place 10 balls (indistinguishable) into eight bins equals the number of 10-combinations from a set with eight elements when repetition is allowed.

$$C(n + r - 1, n - 1)$$

Consequently, there are $C(8+10-1, 10) = C(17, 10) = 17!/10!7! = 19448$.

This means that there are $C(n + r - 1, n - 1)$ ways to place r indistinguishable objects into n distinguishable boxes.

Distinguishable Objects and Indistinguishable Boxes

EXAMPLE: How many ways are there to put four different employees into three indistinguishable offices, when each office can contain any number of employees?

Solution: 14 ways!

We represent the four employees by A, B, C, and D.

- {{A, B,C,D}}. C(4, 4)=1
- {{A, B,C}, {D}}, {{A, B,D}, {C}}, {{A,C,D}, {B}}, {{B, C,D}, {A}}. C(4, 3)=4
- {{A, B}, {C,D}}, {{A,C}, {B,D}}, {{A,D}, {B, C}}. C(4, 2)/2=3
- {{A, B}, {C}, {D}}, {{A,C}, {B}, {D}}, {{A,D}, {B}, {C}}, {{B, C}, {A}, {D}}, {{B,D}}, {A}, {C}}, {{C,D}, {A}, {B}}. C(4, 2)=6

Indistinguishable Objects and Indistinguishable Boxes

EXAMPLE: How many ways are there to pack six copies of the same book into four identical boxes, where a box can contain as many as six books?

Solution: We will enumerate all ways to pack the books.

1) 6
2) 5, 1
3) 4, 2
4) 4, 1, 1
5) 3, 3
6) 3, 2, 1
7) 3, 1, 1, 1
8) 2, 2, 2
9) 2, 2, 1, 1.

Summary: Objects into Boxes

distinguishable boxes

- distinguishable objects: The number of ways to distribute n distinguishable objects into k distinguishable boxes so that n_i objects are placed into box i, i = 1, 2,…, k, equals $\frac{n!}{n_1!n_2!\cdots n_k!}$.

- indistinguishable objects: This means that there are C(n + r − 1, n − 1) ways to place r indistinguishable objects into n distinguishable boxes.

indistinguishable boxes

- No simple closed formula exists for this number.

Section Summary

Permutations with Repetition

Combinations with Repetition

Permutations with Indistinguishable Objects

Distributing Objects into Boxes

Generating Permutations and Combinations

Section 6.6

6.6.1 Introduction

Sometimes permutations or combinations need to be generated, not just counted.

1. Suppose that a salesperson must visit six different cities. In which order should these cities be visited to minimize total travel time?
   - ✓ One way to determine the best order is to determine the travel time for each of the $6! = 720$ different orders in which the cities can be visited and choose the one with the smallest travel time.

2. Suppose we are given a set of six positive integers and wish to find a subset of them that has 100 as their sum, if such a subset exists.
   - ✓ One way to find these numbers is to generate all $2^6 = 64$ subsets and check the sum of their elements.

3. Suppose a laboratory has 95 employees. A group of 12 of these employees with a particular set of 25 skills is needed for a project. (Each employee can have one or more of these skills.)
   - ✓ One way to find such a set of employees is to generate all sets of 12 of these employees and check whether they have the desired skills.

6.6.2 Generating Permutations

Any set with *n* elements can be placed in one-to-one correspondence with the set {1, 2, 3,…, *n*}.

We can list the permutations of any set of *n* elements by generating the permutations of the *n* smallest positive integers ({1, 2, 3,…, *n*}),

and then replacing these integers with the corresponding elements.

We will describe one of these that is based on the lexicographic (or dictionary) ordering of the set of permutations of {1, 2, 3,…, n}.

Permutation in Lexicographic Order

In this ordering, the permutation $a_1a_2\cdots a_n$ precedes the permutation of $b_1b_2\cdots b_n$, if for some k, with $1 \le k \le n$, $a_1 = b_1$, $a_2 = b_2$,…, $a_{k-1} = b_{k-1}$, and $a_k < b_k$.

EXAMPLE: The permutation 23415 of the set {1, 2, 3, 4, 5} precedes the permutation 23514.

Similarly, the permutation 41532 precedes 52143.

Permutation in Lexicographic Order

EXAMPLE: What is the next permutation in lexicographic order after 362541?

362541

Solution:

364125

The last pair of integers a_j and a_{j+1} where $a_j < a_{j+1}$ is $a_3 = 2$ and $a_4 = 5$.

The least integer to the right of 2 that is greater than 2 in the permutation is $a_5 = 4$.

Hence, 4 is placed in the third position.

Then the integers 2, 5, and 1 are placed in order in the last three positions, giving 125 as the last three positions of the permutation.

Hence, the next permutation is 364125.

Permutation in Lexicographic Order

EXAMPLE: Generate the permutations of the integers 1, 2, 3 in lexicographic order?

Solution: Begin with 123.

- ✓ The next permutation is obtained by interchanging 3 and 2 to obtain 132.
- ✓ Next, because 3 > 2 and 1 < 3, permute the three integers in 132.
- ✓ Put the smaller of 3 and 2 in the first position, and then put 1 and 3 in increasing order in positions 2 and 3 to obtain 213.
- ✓ This is followed by 231, obtained by interchanging 1 and 3, because 1 < 3.
- ✓ The next larger permutation has 3 in the first position, followed by 1 and 2 in increasing order, namely, 312.
- ✓ Finally, interchange 1 and 2 to obtain the last permutation, 321.

They are 123, 132, 213, 231, 312, and 321.

Generating the Next Permutation in Lexicographic Order

procedure *next permutation*($a_1 a_2 \ldots a_n$: permutation of
$\{1, 2, \ldots, n\}$ not equal to $n \;\; n-1 \;\; \ldots \;\; 2 \;\; 1$)

$j := n - 1$

while $a_j > a_{j+1}$

  $j := j - 1$

{j is the largest subscript with $a_j < a_{j+1}$}

$k := n$

while $a_j > a_k$

  $k := k - 1$

{a_k is the smallest integer greater than a_j to the right of a_j}

interchange a_j and a_k

$r := n$

$s := j + 1$

while $r > s$

  interchange a_r and a_s

  $r := r - 1$

  $s := s + 1$

{this puts the tail end of the permutation after the jth position in increasing order}

{$a_1 a_2 \ldots a_n$ is now the next permutation}

6.6.3 Generating Combinations

How can we generate all the combinations of the elements of a finite set?

Because a combination is just a subset, we can use the correspondence between subsets of $\{a_1, a_2, \ldots, a_n\}$ and bit strings of length n.

Recall that the bit string corresponding to a subset has a 1 in position k if a_k is in the subset, and has a 0 in this position if a_k is not in the subset.

Recall that a bit string of length n is also the binary expansion of an integer between 0 and $2^n - 1$. The 2^n bit strings can be listed in order of their increasing size as integers in their binary expansions.

To produce all binary expansions of length n, start with the bit string 000…00, with n zeros.

Then, successively find the next expansion until the bit string 111…11 is obtained.

Generating the Next Larger Bit String

How can we generate all the combinations of the elements of a finite set?

At each stage the next binary expansion is found by locating the first position from the right that is not a 1, then changing all the 1s to the right of this position to 0s and making this first 0 (from the right) a 1.

EXAMPLE: Find the next bit string after 10 0010 0111.

Solution:

The first bit from the right that is not a 1 is the fourth bit from the right.

10 0010 0111

Change this bit to a 1 and change all the following bits to 0s.

10 0010 1000

Generating the Next Larger Bit String

procedure *next bit string*($b_{n-1}\, b_{n-2} \ldots b_1 b_0$: bit string not equal to $11 \ldots 11$)
$i := 0$
while $b_i = 1$
  $b_i := 0$
  $i := i + 1$
$b_i := 1$
$\{b_{n-1}\, b_{n-2} \ldots b_1 b_0$ is now the next bit string$\}$

Generating the Next Larger Bit String

Next, an algorithm for generating the r-combinations of the set $\{1, 2, 3, \ldots, n\}$ will be given.

An r-combination can be represented by a sequence containing the elements in the subset in increasing order. The r-combinations can be listed using lexicographic order on these sequences.

In this lexicographic ordering, the first r-combination is $\{1, 2, \ldots, r-1, r\}$ and the last r-combination is $\{n-r+1, n-r+2, \ldots, n-1, n\}$.

The next r-combination after $a_1 a_2 \cdots a_r$ can be obtained in the following way:

- ✓ First, locate the last element a_i in the sequence such that $a_i \neq n - r + i$.
- ✓ Then, replace a_i with $a_i + 1$ and a_j with $a_i + j - i + 1$, for $j = i + 1, i + 2, \ldots, r$.

Generating the Next Larger Bit String

EXAMPLE: Find the next larger 4-combination of the set {1, 2, 3, 4, 5, 6} after {1, 2, 5, 6}.

1, 2, 5, 6

Solution:

1, 3, 4, 5

The last term among the terms a_i with $a_1 = 1$, $a_2 = 2$, $a_3 = 5$, and $a_4 = 6$ such that $a_i \neq 6 - 4 + i$ is $a_2 = 2$.

To obtain the next larger 4-combination, increment a_2 by 1 to obtain $a_2 = 3$.

Then set $a_3 = 3 + 1 = 4$ and $a_4 = 3 + 2 = 5$.

Hence the next larger 4-combination is {1, 3, 4, 5}.

Generating the Next r-Combination in Lexicographic Order

procedure *next r-combination*($\{a_1, a_2, \ldots, a_r\}$: proper subset of $\{1, 2, \ldots, n\}$ not equal to $\{n - r + 1, \ldots, n\}$ with $a_1 < a_2 < \cdots < a_r$)

$i := r$

while $a_i = n - r + i$

  $i := i - 1$

$a_i := a_i + 1$

for $j := i + 1$ **to** r

  $a_j := a_i + j - i$

$\{\{a_1, a_2, \ldots, a_r\}$ is now the next combination$\}$

Section Summary

Generating Permutations

Generating Combinations

Chapter Summary

6.1 The Basics of Counting

6.2 The Pigeonhole Principle

6.3 Permutations and Combinations

6.4 Binomial Coefficients and Identities

6.5 Generalized Permutations and Combinations

6.6 Generating Permutations and Combinations

Key Terms and Results

TERMS

combinatorics: the study of arrangements of objects

enumeration: the counting of arrangements of objects

tree diagram: a diagram made up of a root, branches leaving the root, and other branches leaving some of the endpoints of branches

permutation: an ordered arrangement of the elements of a set

***r*-permutation:** an ordered arrangement of r elements of a set

$\boldsymbol{P(n,r)}$**:** the number of r-permutations of a set with n elements

***r*-combination:** an unordered selection of r elements of a set

$\boldsymbol{C(n,r)}$**:** the number of r-combinations of a set with n elements

binomial coefficient $\binom{n}{r}$**:** also the number of r-combinations of a set with n elements

combinatorial proof: a proof that uses counting arguments rather than algebraic manipulation to prove a result

Pascal's triangle: a representation of the binomial coefficients where the ith row of the triangle contains $\binom{i}{j}$ for $j = 0, 1, 2, \ldots, i$

$\boldsymbol{S(n,j)}$**:** the Stirling number of the second kind denoting the number of ways to distribute n distinguishable objects into j indistinguishable boxes so that no box is empty

Key Terms and Results

RESULTS

product rule for counting: The number of ways to do a procedure that consists of two tasks is the product of the number of ways to do the first task and the number of ways to do the second task after the first task has been done.

product rule for sets: The number of elements in the Cartesian product of finite sets is the product of the number of elements in each set.

sum rule for counting: The number of ways to do a task in one of two ways is the sum of the number of ways to do these tasks if they cannot be done simultaneously.

sum rule for sets: The number of elements in the union of pairwise disjoint finite sets is the sum of the numbers of elements in these sets.

subtraction rule for counting or **inclusion–exclusion for sets:** If a task can be done in either n_1 ways or n_2 ways, then the number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways.

subtraction rule or **inclusion–exclusion for sets:** The number of elements in the union of two sets is the sum of the number of elements in these sets minus the number of elements in their intersection.

division rule for counting: There are n/d ways to do a task if it can be done using a procedure that can be carried out in n ways, and for every way w, exactly d of the n ways correspond to way w.

division rule for sets: Suppose that a finite set A is the union of n disjoint subsets each with d elements. Then $n = |A|/d$.

the pigeonhole principle: When more than k objects are placed in k boxes, there must be a box containing more than one object.

the generalized pigeonhole principle: When N objects are placed in k boxes, there must be a box containing at least $\lceil N/k \rceil$ objects.

$$P(n, r) = \frac{n!}{(n-r)!}$$

$$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Pascal's identity: $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$

the binomial theorem: $(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$

There are n^r r-permutations of a set with n elements when repetition is allowed.

There are $C(n+r-1, r)$ r-combinations of a set with n elements when repetition is allowed.

There are $n!/(n_1!\, n_2! \cdots n_k!)$ permutations of n objects of k types where there are n_i indistinguishable objects of type i for $i = 1, 2, 3, \ldots, k$.

the algorithm for generating the permutations of the set $\{1, 2, \ldots, n\}$